

# MANASA BANDLA

+91 8341820801 ✦ Hyderabad, India

[manasabandla66@gmail.com](mailto:manasabandla66@gmail.com) ✦ [linkedin.com/in/manasa-bandla-513407226](https://www.linkedin.com/in/manasa-bandla-513407226)

## PROFILE SUMMARY

Data Scientist with 2+ years of experience applying machine learning, artificial intelligence, and statistical modeling to solve complex business problems using Python and SQL. Expertise in developing predictive models for classification and regression, performing feature engineering, and conducting exploratory data analysis (EDA) to generate actionable insights. Experienced in managing end-to-end machine learning initiatives including data collection, data cleaning, preprocessing, model development, evaluation, and deployment. Skilled in building Retrieval-Augmented Generation (RAG) pipelines and agentic AI solutions using LangChain, with strong proficiency in data visualization using Matplotlib, Seaborn, and Power BI. Strong ability to align data science solutions with business objectives and support strategic decision making.

## EDUCATION

**Bachelor's Degree in Electrical and Electronics Engineering**, G. Pullareddy College 2019 – 2023

## SKILLS

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| <b>Programming &amp; Data</b>    | Python, R, SQL, Power BI, NumPy, Pandas, Matplotlib, Seaborn<br>Data Collection, Data Cleaning & Preprocessing, Data Visualization<br>Feature Engineering, Exploratory Data Analysis (EDA)                                                                                                                                                                                                                                            |
| <b>Machine Learning &amp; AI</b> | Supervised & Unsupervised Learning, Regression, Classification, Clustering<br>Statistical Modeling, Hypothesis Testing, Linear & Multivariate Regression<br>Time Series Forecasting, Analytical Modelling<br>Model Training, Cross-Validation, Hyperparameter Optimization<br>Linear Regression, Logistic Regression, Decision Trees, Random Forest<br>AdaBoost, XGBoost, LightGBM, Scikit-learn<br>PySpark, PyTorch, TensorFlow, NLP |
| <b>Generative AI &amp; LLMs</b>  | Retrieval-Augmented Generation (RAG), Semantic Search<br>LangChain, Agentic AI Solutions, Prompt Engineering, AI Agents, MCP<br>LLMs: ChatGPT (OpenAI)                                                                                                                                                                                                                                                                                |
| <b>Tools &amp; Technologies</b>  | MySQL, Git, Azure (AZ-900), VS Code, Postman, Ubuntu/Linux                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Soft Skills</b>               | Strong Communication, Problem Solving & Analytical Thinking<br>Attention to Detail, Team Collaboration, Learning Agility<br>Strategic Thinking & Business Alignment                                                                                                                                                                                                                                                                   |

## EXPERIENCE

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| <b>Data Scientist</b><br>IBM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | May 2024 – Present<br><i>Hyderabad, India</i> |
| <ul style="list-style-type: none"><li>Developed end-to-end data science initiatives including data collection, cleaning, preprocessing, and predictive modeling for regression and classification problems using Python and SQL, aligning technical solutions with business objectives.</li><li>Performed exploratory data analysis (EDA) and feature engineering to uncover patterns and improve model performance, communicating findings through data visualization using Matplotlib and Power BI.</li><li>Applied advanced machine learning expertise including boosting algorithms (XGBoost, LightGBM), decision trees, and time series modeling to address key business challenges using scikit-learn.</li><li>Implemented model optimization using cross-validation and hyperparameter tuning, applying statistical concepts including hypothesis testing to ensure robust performance and scalability.</li><li>Developed monitoring and evaluation workflows to track model performance and deliver interpretable insights to stakeholders in Agile environments using Git for version control.</li></ul> |                                               |

PROJECTS

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CIBC Credit Card Churn Prediction

**Description:** Developed a machine learning pipeline to predict credit card customer churn for CIBC (Canadian Imperial Bank of Commerce), enabling the bank to identify high-risk customers and implement targeted retention strategies using behavioral and transactional data.

Responsibilities:

- Built a predictive churn model using Python and SQL on a dataset of 7,000+ customers; performed data collection, data cleaning, and preprocessing to ensure data quality and integrity.
- Conducted exploratory data analysis (EDA) and feature engineering to identify trends, patterns, and key predictors influencing customer churn behavior.
- Trained and optimized classification models including Logistic Regression, Decision Trees, Random Forest, and XGBoost using scikit-learn, achieving 89% prediction accuracy.
- Evaluated model performance using Precision, Recall, F1-score, and ROC-AUC, applying statistical concepts including hypothesis testing to validate results (ROC-AUC: 0.91).
- Generated explainable insights using SHAP and data visualization techniques (Matplotlib, Seaborn) to identify key factors influencing customer churn; documented processes and findings clearly for stakeholders.

Enterprise Knowledge Assistant using RAG and LLMs

**Description:** Developed a Retrieval-Augmented Generation (RAG) based enterprise knowledge assistant that enables users to query internal documents and retrieve context-aware answers using large language models, leveraging semantic search for accurate information retrieval.

Responsibilities:

- Built an end-to-end RAG pipeline using Python and LangChain to enable semantic search across enterprise documents, developing an agentic AI solution for intelligent document querying.
- Implemented document ingestion, chunking, and embedding generation using transformer-based embedding models to support efficient information retrieval.
- Stored vector embeddings in a Chroma vector database to enable efficient similarity-based semantic search and retrieval.
- Integrated LLMs (ChatGPT/OpenAI) with the retrieval pipeline to generate context-aware answers from retrieved document chunks, reducing hallucinations through prompt engineering.
- Maintained clear and concise documentation of the RAG pipeline architecture, data processes, and analytical findings for knowledge transfer and reproducibility.

CERTIFICATIONS

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- **Microsoft Certified:** Azure Fundamentals (AZ-900)
- **IBM Skills Build:** Artificial Intelligence Fundamentals

ADDITIONAL INFORMATION

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| Nationality | Indian                 |
| Languages   | English, Hindi, Telugu |